

STATISTICS & PROBABILITY NOTES

BY SHAHZAIB BAIG

Complete verbatim course notes compiled into a high-contrast, print-ready booklet.
Every word, table, hand-drawn diagram, and formula is taken directly from the original
notes — beautifully formatted for exam revision.

```
► terminal
$ whoami
shahzaib_baig

$ git remote -v
origin  https://github.com/Shahzaib695/Ai-Data-Science.git

$ wc -l notes.odt
557 elements · 105 diagrams · A4 booklet
```

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FORMULA REFERENCE SHEET

MEAN

$$\bar{x} = \frac{\sum x_i}{n}$$

Sum divided by count

RANGE

$$R = R_{\max} - R_{\min}$$

Max minus minimum

IQR

$$IQR = Q_3 - Q_1$$

Middle 50% spread

FENCES

$$UF = Q_3 + 1.5IQR, LF = Q_1 - 1.5IQR$$

Outlier boundaries

POP. VARIANCE

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Avg squared deviation

SAMP. VARIANCE

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Bessel's correction

Z-SCORE

$$Z = \frac{x - \mu}{\sigma}$$

Std deviations from mean

PROBABILITY

$$P(A) = \frac{Favourable}{Total}$$

Basic probability

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

A or B

MULTIPLICATION

$$P(A \cap B) = P(A) \cdot P(B|A)$$

A and B

CONDITIONAL

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A given B

BAYES'

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Update prior with evidence

Z-TEST

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Known population σ

T-TEST

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}, df = n - 1$$

Unknown σ

CHI-SQUARE

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Categorical association

COVARIANCE

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Joint variance

CORRELATION

$$r = \frac{\text{Cov}(X, Y)}{s_x \cdot s_y}$$

-1 to +1

P-VALUE

$$p = 2 \cdot \min(P(Z))$$

Two-tailed coding

STATISTICS AND PROBABILITY:

DATA:

A set of information is called Data. For every operation of analysis, machine learning, deep learning and vice versa we need data to perform operations.

HOW TO STORE DATA:

DATA TABLE:

We store data in form of tables which is known as Data Table.

We also have rows and columns in this data.

Columns

Rows

ONE WAY TABLE:

EXAMPLE

Suppose there are 2 columns one is Name and other is Age. Here we can say Name is an individual and age is a variable because individual has dependencies on multiple things.

Here in the above table u can see name is acting as individual and others are acting as like variables because when someone asks age about a particular person he/she has to ask it using the name here name act as an independent individual while others act as a variable.

The variables that are divided based on the category we call them Qualitative Variables while which are not divided in any category we call them as Quantitative Variables. Here in the above example gender is qualitative variable while age and salary are quantitative variables.

A more proper definition of both are as follows:

Continuous numeric data like age, salary, height and etc are known as Quantitative Variables.

Where as categorical data like gender, genre and etc are known as Qualitative Variables.

DATA VISUALIZATION

BAR GRAPH:

Its the simplest and the most common graph the thing is that if we have data of the most liked fruit by people to manually verify it we need to count it but in statistic we can do it by visualizing a graph.

To create a bar graph first we need to create a frequency table for fruits.

After finding the frequency of the fruit or the count then we can draw horizontal lines each line tells the specific frequency of the fruit we can then easily plot the graph. The underline approach and the plotted graph image are as follows:

PERSONS	FRUITS
A	Apple
B	Banana
C	Apple
D	Apple
E	Banana
F	Banana
G	Apple
H	Kiwi
I	Kiwi
J	Apple
K	Mango
L	Mango
M	Apple
N	Banana

FRUITS	COUNT
Apple	6
Banana	4
Kiwi	2
Mango	2

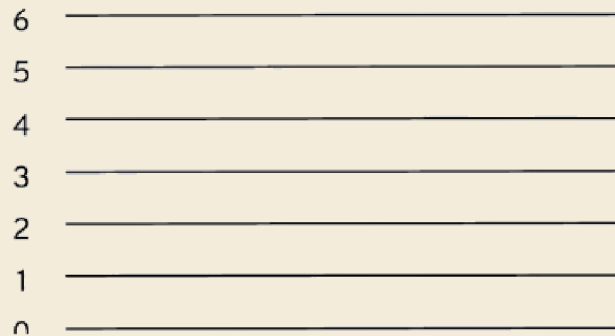
So here you got the frequency table of fruits.



Now I again ask the questions.

Which is most liked fruit ?

Which is the least liked fruit ?



PIE CHART:

Now lets see pie chart it uses the frequency approach just like the bar graph but here we use the percentage occurrence approach.

Lets take a look on the frequency of the fruits data lets see that but the twist is that we will be converting those numeric data into percentage.

FORMULA

Formula → $\frac{\text{No. Of Frequency}}{\text{Total No Of Fruits}} \times 100$ Frequency

Mathematical calculations are done in the register please refer to that

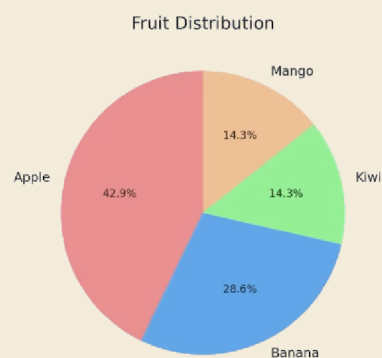
Apple → 42.9%

Banana → 28.6%

Kiwi → 14.3%

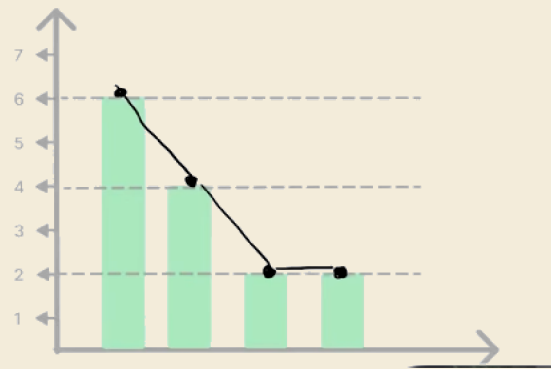
Mango → 14.3%

For creation of the pie chart in the back-end a circle is created then it is divided with the percentage data that we have calculated and ultimately something like this is made:

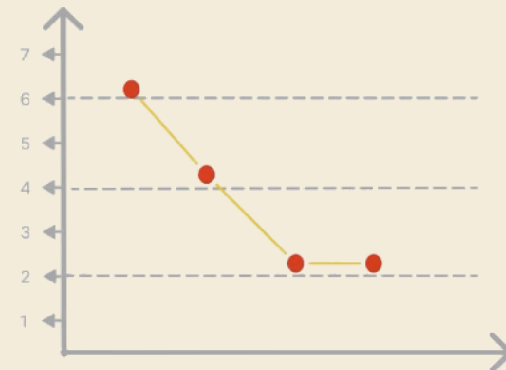


LINE GRAPH:

Line graph are interesting but for making it we need to look backwards towards bar graph.



For understanding lets look at the bar graph we created now lets add dots on the top of the bars and connect them.



After that remove the bars boom here we got our very own line graph.

QUESTION

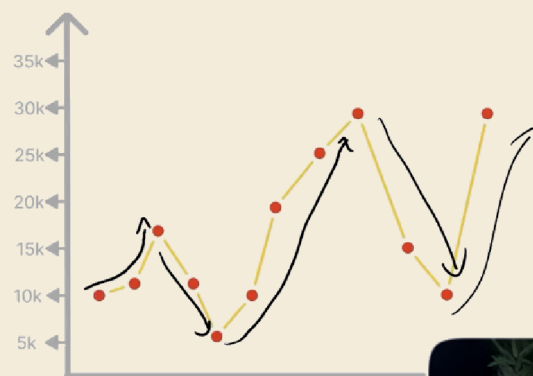
Question: Now when to use bar graph and when to use line graph??

Answer: To explain this we will be using a data of monthly expenses.

Here u can see we cant calculate the frequency of the data here but we can only plot a simple bar graph which cant show the flow. In short we will use bar graph when we have frequency of data and we will be using line graph when we have to see a trend like here I am attaching the picture of the line graph that will be made for this data which is attached on the next page.

Data Line Graph

Month	Expense
Jan	10k
Feb	12k
Mar	18k
Apr	12k
May	5k
Jun	10k
Jul	20k
Aug	25k
Sep	30k
Oct	15k
Nov	10k
Dec	30k



Here we can see it is showing the flow of trend like the expenses flow from month 1 to month 12 this is the core reason to create a line graph.

O-GIVES:

To create an o-gives we have to take out cumulative frequencies.

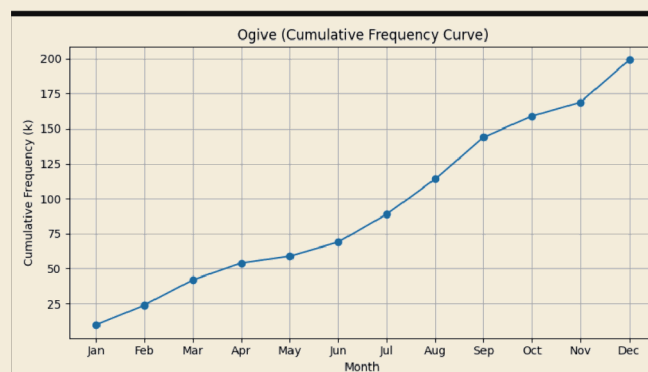
So now what is cumulative frequencies lets calculate what is it:

Cumulative Frequencies → I have attached the table below now the situation is that we create 2 tables one is of the default expenses and in the next table we will be adding per month expense like in Jan we have 10k expenses and for Feb we have 12k expenses and for march we have 20k expenses now we have the data for the cumulative frequency table we will be doing like this. For Jan it stays the same as 10k and for Feb it will now be 22k (10k of Jan + 12k of Feb) and for march it will now be 42k (10k of Jan + 12k of Feb + 20k of Mar).

Month	Expense
Jan	10k
Feb	12k
mar	18k
apr	12k
may	5k
jun	10k
jul	20k
aug	25k
sep	30k
oct	15k
nov	10k
dec	30k

Month	Expense
Jan	10k
Feb	22k
mar	42k
apr	54k
may	59k
jun	69k
jul	89k
aug	114k
sep	144k
oct	159k
nov	169k
dec	199k

Now the line graph that will be plotted on the basis of cumulative frequencies will be known as O-Gives.



Which is drawn here →

TWO WAY TABLE:

Currently the table we have on the right is a one way table. Lets assume this table is of data of the period of revenue of 4 years like from 2020-2023 now we can break it down as follows because we don't have separate data of each year we only have a combined data so we will break it down as follows:

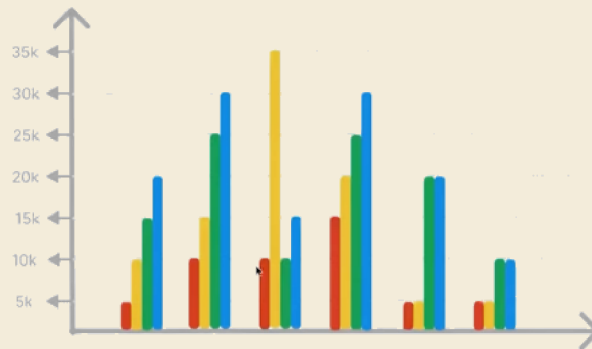
Month	revenue
Jan	50k
mar	80k
may	70k
jul	90k
sep	50k
nov	30k

Month	2020	2021	2022	2023	Total
Jan	5k	10k	15k	20k	50k
mar	10k	15k	25k	30k	80k
may	10k	35k	10k	15k	70k
jul	15k	20k	25k	30k	90k
sep	5k	5k	20k	20k	50k
nov	5k	5k	10k	10k	30k
Total	50k	90k	105k	125k	370k

now this is a two way table because now month is not only the independent thing like if someone asks a revenue from us we will be asking two questions like for which month and which year. Dependencies has switched from a single month to a year too. Now if

someone asks us when was the revenue 35k we can tell them it was in the year 2021 in the month of May.

Now using this data we can create a line graph for each year. Or even a bar graph for each year

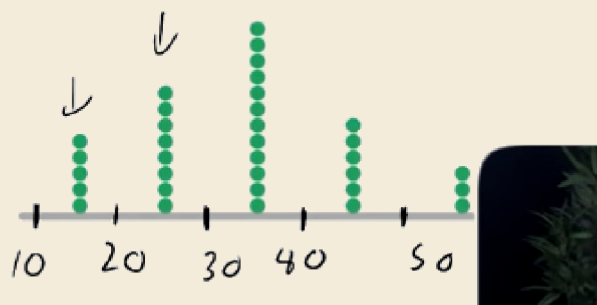


In 2 way table there is a thing known as Dot Plot its basically a bar graph with dots stacked on each other. To create that we basically need frequency of specific data which fulfill the need.

Which needs data like this →

It helps to build understanding for histogram.

Age Group	Members
10 - 20	5
20 - 30	8
30 - 40	12
40 - 50	6
50 - 60	3



Dot Plot of the data

RELATIVE FREQUENCY TABLE:

A relative frequency table can be made through both one way and two way table but most of the data we see around is two way so we will focus on that

Data →

	Red	White
Rose	20	50
Lily	5	30

(its a two way table because there is dependencies on two things the color and the flower like if someone asks us where is 20 so u will be telling them in red column and rose rows something like that I hope u got my point)

Now lets calculate the column frequency of the data for which we will be calculating the frequency from 100. (For detailed calculations refer to the notebook we have done calculations there)

the white ones are different its due to a mathematical error u can see the notebook for instance.

Column frequency	Red	White
Rose	0.80	0.71
Lily	0.20	0.29

FORMULA

Formula for calculating → No. of An Element * 100

Total Number

We have calculated the column frequency so if we total the column frequency we will be getting 100% but not for rows. Fortunately we can also calculate row related frequencies.

We do this because in Machine Learning We Mostly need to convert numeric data into 0s and 1

JOINT DISTRIBUTION:

A joint distribution table is similar to relative frequency table as it also contains percentage. This table compares two different distributions in a single table.

So we can say if we join 2 way tables and we use percentage and total percentage comes at 100% this is called as joint distribution.

There you go this is a joint distribution as there is a distribution for weight loss and distribution of total hours workout and we are comparing both the distribution so JOINT DISTRIBUTION 😊

		Weight Lost			
		0-2	2-4	4-6	
Total hours workout	3-6	13%	5%	7%	25%
	6-9	5%	15%	10%	30%
	9-12	7%	13%	25%	45%
		25%	33%	42%	100%

HISTOGRAM:

Histogram looks actually like a bar graph but here we group the data into buckets or bins.

Candidates	age
p1	21
p2	22
p3	45
p4	56

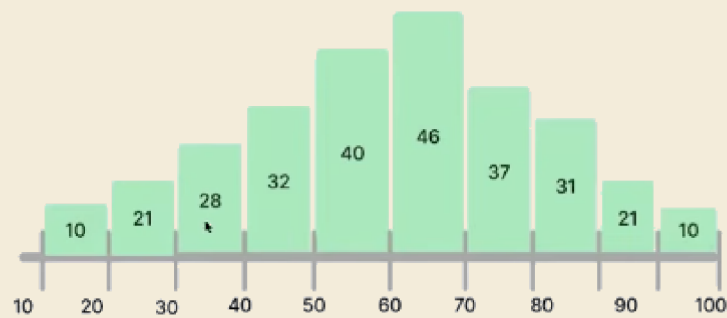
We will be seeing using this data →

This data is incomplete it has multiple rows but for adding this here we are using only this data.

Now for making histogram we will be using x-axis and we will be creating bins just like this:
these bins are created using range function :).



Now in these bins we will put the candidate into their desired bins using their age. Just like the image I have attached →



The more the number of data in the bins the more tall the histogram becomes

Now let's work on actual data

Now let's calculate the range of this to calculate the range we use the formula $\rightarrow \text{Range} = R_{\text{max}} - R_{\text{min}}$

There are several persons going to gym and the hours they are working are

3,4,4,5,5,5,6,6,7,8,9,10,10,10,10,11,11,12,13,14,15,15,18,18,20,20,20,21,24,24.

here max is 24 and min is 3 so range is 21

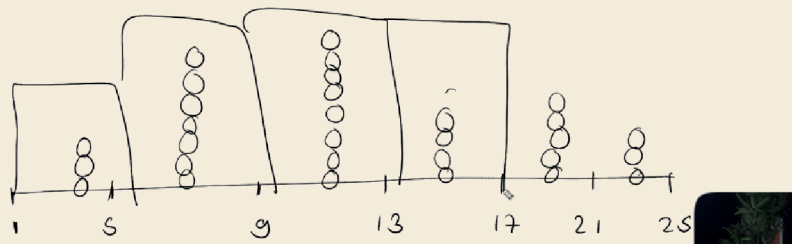
For creating bins we usually have to play around like here we can do is divide the range with 6 $21/6 = 3.5$.

For safe side we will round it off to 4 and the bins we will have is.

1-5, 6-10, 11-15, 16-20, 21-25

3,4,4,5,5,5,6,6,7,8,9,10,10,10,10,11,11,12,13,14,15,15,18,18,20,20,20,21,24,24.

So our bins are - 1-5, 5-9, 9-13, 13-17, 17-21, 21-25.



QUESTION

Question: Why and When we use Histogram?

Answer: We use Histogram when we have quantitative data, and we use it to group that data into buckets called bins so we can see the shape or spread at a glance like whether most values are clustered together, spread out evenly, or piled up on one side instead of having to read through a long list of raw numbers to figure that out.

DATA ANALYZING

In this section we will be focusing how we can do things mathematically to better understand our data

MEASURE OF CENTRAL TENDENCY:

It means it is the way we can describe the middle part of our data. The importance of middle part of the data we will see it further in the course why is it this much important.

Mean $\rightarrow$ The first way to find the middle part of the data is mean. It is also known as average.

EXAMPLE

Example $\rightarrow$ Here we have data 1,2,3,4,5. Here 3 is the middle part because its continuous data like equal spacing between elements.

But when it comes to non continuous data let's see it. Here we have data 2,2,7,9,10.

FORMULA

Formula $\rightarrow \text{Sum of all numbers} / \text{total numbers}$

Mean = 6 for the non continuous data refer to the notebook for detailed calculation

Fancy representation of this formula is statistic $\rightarrow$

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Details of this are mentioned in notebook

Median $\rightarrow$ The second way to find the middle part of the data. It doesn't necessarily need to be calculated through mathematical formulae it is simple.

EXAMPLE

Example $\rightarrow$ 4,7,9,10 The middle part of this data is 7 and 9 so we will add them both and divide them by 2 we will get 8 and 8 will be the middle point of this data.

Another example 1,2,6,7,9 no need for calculation here we can exactly see that 6 is the midpoint like on the left of 6 we have 2 elements and on the right too we have 2 elements so 6 automatically becomes the midpoint.

Mode $\rightarrow$ It is the most simplest one. Mode is the most repeated element in the data.

EXAMPLE

Example $\rightarrow$ 2,2,5,6,7,8,10,10,10 Here we can clearly see 10 is the most repeated value in the data hence 10 is the mode of this data.

Now for suppose we have data 1,2,4,5 we can not take out mode from this as there is no repeated value in this data. On the other hand we have this type of data 1,1,1,2,3,4,5,5,5,6,7 in this data we have 2 repeated values which are 1 and 5 so this becomes a case of bimodal even if there are 4 and 5 number of modes this lies in the case of bimodal.

MEASURE OF SPREAD:

It means how much and how are data spread around its center. Its different from what we have seen above.

It means in simple words how much your data is spread from the central point.

We will be talking about Range, IQR (Interquartile range) and Standard Deviation as they are also measure of spread. But here we will be covering Range and IQR for now and variance and std deviation are part too but we will discuss them later.

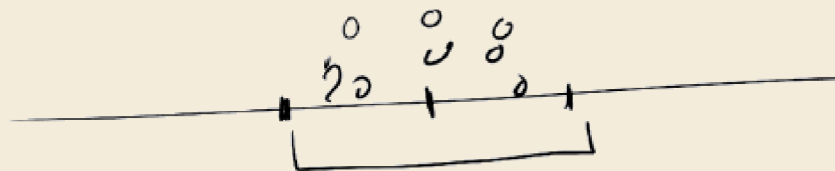
RANGE:

To find the range we subtract the maximum value in the data with the minimum value in the data doing this we get Range.

FORMULA

Formula $\rightarrow$ Range = Rmax - Rmin.

Range tells us the distribution of data how and in what range they are distributed.



Refer to the diagram suppose u have data and in the center u have the mean or your center data and on the right and left u have ur highest value and the lowest value we can see how close or how far these values are to check how spreaded the data is between them.

INTERQUARTILE RANGE(IQR):

Before moving towards Interquartile we first need to understand what are Quartiles. Because in IQR we will be measuring the distance between one quarter and another quarter of our data.

There are mostly 3 types of percentile and sometimes 4 types of percentile.

Q1-25% Percentile

Q2-50% Percentile

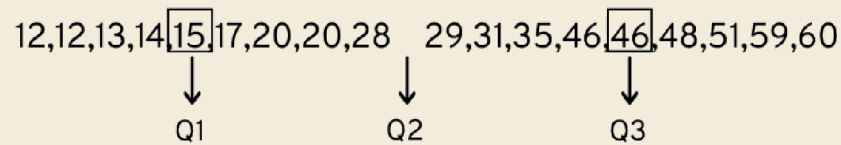
Q3-75% Percentile

Q4-100% Percentile

Percentile means the percentage of data that lies below a specific value.

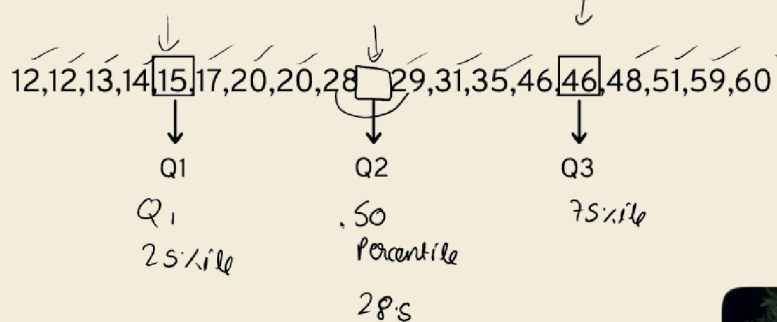
To understand this better let see a data we have a data

12,12,13,14,15,17,20,20,28,29,31,35,46,46,48,51,59,60



We can calculate Q1,Q2,Q3 by dividing the data into two parts like in the data we have 18 elements we can divide them into 9 9 parts for which then we can calculate the data. For the calculating Q1 we can see the 5th value of the data that will be Q1 for the Q2 we can do is divide the 9th and 10th element and we can get Q2 which exactly means the mean of the data. And for Q3 we will do the same thing but with the second part of the data choosing the 5th value of the data it will be Q3.

So after finding the percentile we get.



Now to find IQR we just have to subtract Q1 From Q3 $46-15 = 31$.

So if the IQR is low it means the data is close to the median if the IQR is high it means the data is highly spread from the center.

OUTLIERS:

To understand outliers we need to look at an example of data first only in that way we can understand what really outliers are

1,2,4,7,9,10,101.

Lets find the central tendency of this data we will be using mean and median since we cant use mode because there are no repeated values.

Mean $\rightarrow 1+2+4+7+9+10+101 = 134/7 \rightarrow 19.4$

Median $\rightarrow 1,2,4,7,9,10,101 = 7$

Here u can see how much big difference there is between the mean and median its due to an outlier (101) and if we remove that mean and median would be very close then.

OUTLIERS DAMAGE OUR DATA BY:

- They distort measures like mean and standard deviation making them less representative to the given data.
- In machine learning especially in regression models outliers can disproportionately influence the model leading to poor generalization.
- Outliers impact our model performance our statistical testing and many other thing so we have to get rid from them.

So now to get rid of outliers we use the upper and lower fence approach in our data by that we can get rid of them.

The points that show linking between them is the point till we consider our data after the end line no data will be considered.



here we will use our IQR, Q1 and Q3 to find the upper and lower fence.

Lets understand with the example.

1, 1, 2, 2, 3, 4, 5, 5, 6, 6, 7, 8, 9, 9, 10, 10, 11, 13, 13, 14, 15, 101

(Mathematical calculation done is notebook)

To find upper and lower fence we have Formula to calculate each fence.

Upper Fence $\rightarrow Q3 + 1.5(IQR)$

Lower Fence $\rightarrow Q1 - 1.5(IQR)$

For this example we got this after calculations

$$IQR = 7$$

$$Q1 = 4$$

$$Q3 = 11$$

Upper Fence $\rightarrow 21.5$

Lower Fence $\rightarrow -6.5$

We have to check whether there is any value below lower fence or any value above the upper fence if those exists they are outliers.

5 Number Summary:

The 5 number summary is prepared after removing outliers from the data it is a statistical summary used to describe the distribution of a dataset. It provides a quick overview of the data's spread and center and it includes the following values:

- Minimum
- First Quartiles (Q1)
- Median (Q2)
- Third Quartiles (Q3)
- Maximum

Lets work with data

2,4,7,10,15,20,25,205 calculations done in notebook

Data after doing calculations

$$Q1 = 5.5, Q2 = 12.5, Q3 = 22.5, IQR = 17, UF = 48, LF = -20$$

Preparing 5 Number Summary After Removing Outliers

Min $\rightarrow 2$

Q1 → 4

Q2 → 10

Q3 → 20

Max → 25

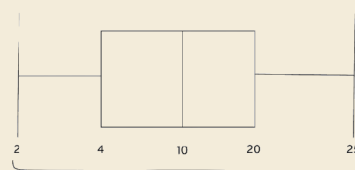
This summary is used for plotting box plots u will understand better when u will see data visualization.

5 NUMBER SUMMARY

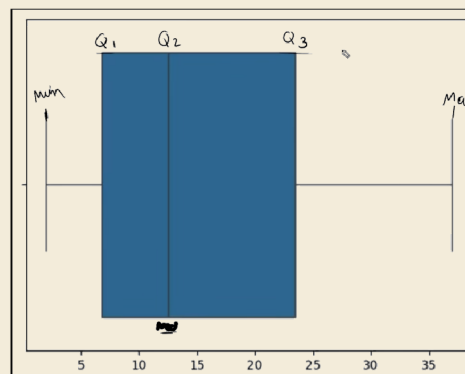
2, 4, 7, 10, 15, 20, 25, 205

- Minimum
- First Qua
- Median (C
- Third Qua
- Maximum

Box Plot



(We have also implemented Code for outliers too see In the repository)

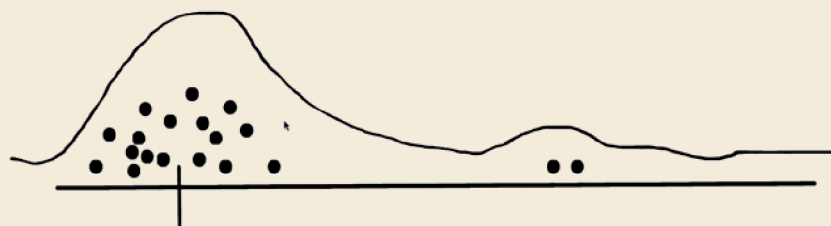


DATA DISTRIBUTION

In this section we will create a base about hypothesis testing.

We will learn about kernel density estimation curve what we draw in histogram. We will also see how the data is spread around the mean.

Lets see with the example of a distribution. In the below distribution we can see how the curve is disturbed because of two outliers that's the sole reason why our bell curve/ kernel density estimation curve is a bit odd.



In this section we will see how data distribution starts with a histogram then it becomes frequency polygon and eventually turns into a density curve.

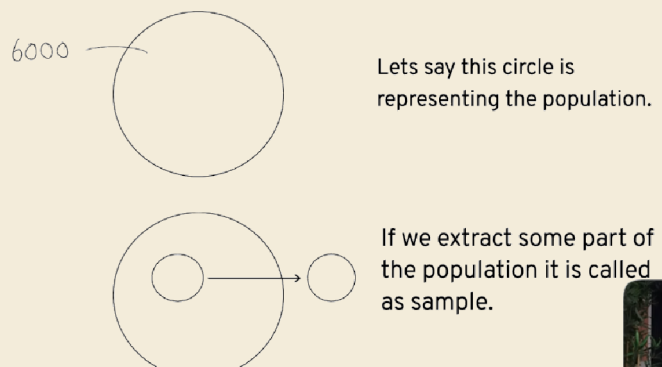
Histogram → Frequency Polygon → Curve

NORMAL DISTRIBUTION:

Distribution which always forms a bell shaped curve. Its used in many areas.

MEAN, VARIANCE AND STANDARD DEVIATION:

Before we begin lets understand what basically is sample and population in stats



To understand properly what a sample is lets take an example of a medicine like we have created a medicine for cancer we have to test it so we can't test on all people so what we will do is gather people from around the globe lets take an example like we take out 500 people from different regions and what will be the result of that test will be considered for the whole population. This is done in Inferential Statistics. We will do it later.

NOTE

Note: When discussing Mean, Variance and Standard Deviation. It is important to clarify that whether we are referring to a population or standard deviation. As formula for each differ.

VARIANCE:

Variance is a statistical measure that quantifies the spread or dispersion of set of data points around their mean.

In short variance tells the value of the data which is clustered around the mean or how the data is spread around the mean.

To find the variance we first see a data a simple one:

85,90,95,100,105

→ Find Mean first = $(85+90+95+100+105)/5 = 95$

→ Then Find the computed difference to find it we will be doing this subtracting the data numbers with mean and wrapping each block with square adding them

$$(85-95)^2 + (90-95)^2 + (95-95)^2 + (100-95)^2 + (105-95)^2 = 250$$

→ Now just divide the computed difference with no of elements and here we can see the value of variance → $250/5 = 50$

50 here is the variance value

← Formula of variance in statistical terms. Lets break it down what each means for instance (This formula is for population).

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

N means no of elements, i=1 means no of elements we will be moving as the elements number increase

g small n. Mean of population is represented by μ and for sample it is represented by $\bar{x}$. And to represent mean of population we will be using sigma symbol with a square and for sample we represent using small s with square. Standard deviation is represented sigma and small s sigma for population and small s for sample

Population	Sample
$\rightarrow N$	$\textcircled{n} \rightarrow \text{Total elements}$
$\rightarrow \mu$	$\bar{X} \rightarrow \text{Mean}$
$\rightarrow \sigma^2$	$S^2 \rightarrow \text{Variance}$
$\rightarrow \sigma$	$S \rightarrow \text{STD}$

(Proper breakdown of each thing is present in the notebook typing here is a challenge for me.)

If we want to calculate variance of sample the formula just becomes this $\rightarrow$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$$

STANDARD DEVIATION:

Standard Deviation is just the square root of the variance its as simple as that. The symbols will be changed depending on for what u r taking standard deviation of is it population or sample.

Basically standard deviation follows the same process u first take out mean then distance (computational difference) then divide it with total number of elements.

85,90,95,100,105

variance = 50

standard deviation $\sqrt{50} = 7.071$

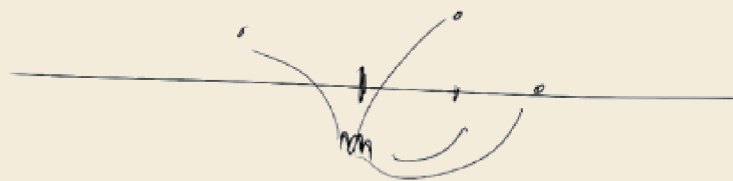
$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

NOTE:

By default use sample formulae unless population or entire or anything related to this is mentioned.

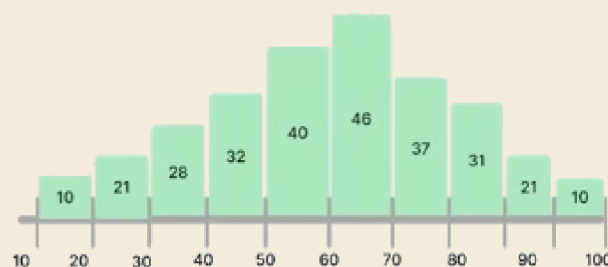
Q: What does it do?

A: Standard deviation tells that how much distance should be between the mean and the points.



DENSITY CURVE:

To understand this concept first we need to understand what is histogram as we already have discussed histogram I will not be discussing here again.



STEPS TO CREATE A DENSITY CURVE:

- First create a histogram just like this $\rightarrow$
- Now we just have to plot points above the histogram and connect them just like we were doing for line graph just like this $\rightarrow$

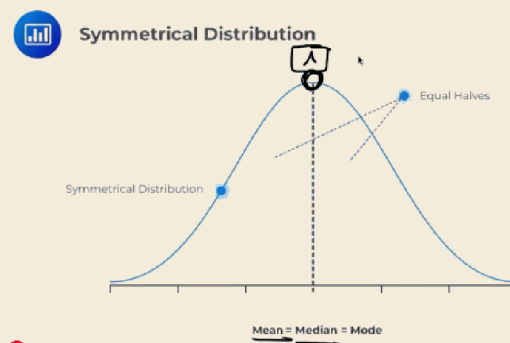
- After this we have to remove the histogram bars. The fewer bin we use the more inaccurate the frequency polygon we will get. Now we have a frequency polygon just like this →
- Now imagine u are using more bins the dots will become more closer and eventually it will become smooth, continuous probability curve which is also known as a density curve.

Now this density curve tells us many things like for this histogram we want to know how much percent of data we have for the people aging between 80-100 we can easily tell that. We will be learning a concept known as z-score from it we can easily tell these types of data. For now we have understood how this curve is made.

TYPES OF CURVES:

SYMMETRICAL DISTRIBUTION:

Symmetrical Distribution also known as Normal Distribution or Gaussian Distribution.



The center line u can see in the picture in this distribution this is the mean median and mode.

Let me explain this concept in the data if the data is normal its obvious the mean or the average value of data would be something that is center of them. Median is already the center of the data. Mode u can see in the picture that the density curve is at its peak in the center so that possibly means the value is repeated in the data that's why it is at its peak. This explains how mean , median and mode lies in the center of this data.

In this distribution u can see both sides right and left sides are equal. This is the thing for this distribution that they are equally distributed in both sides left side and right side.

This distribution always forms a bell shaped curve. We will talk about this distribution in the upcoming chapters.

Right Skewed Distribution And Left Skewed Distribution:

First lets see what skewness is before discussing these distributions.

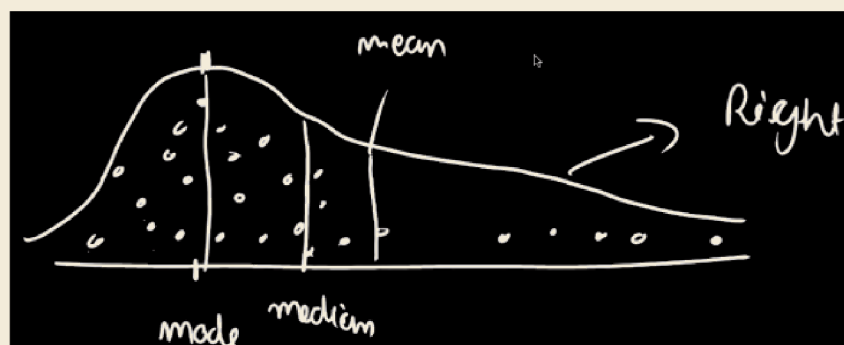
SKEWNESS:

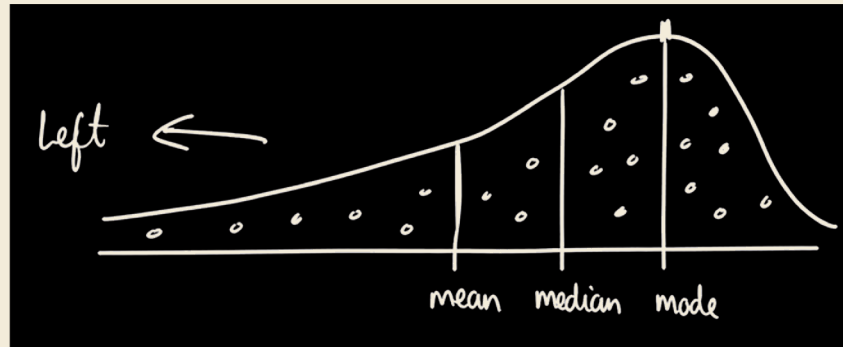
It is the measure of the asymmetry or imbalance in a dataset. In a perfectly balanced data such as normal distribution. The data is perfectly balanced and aligned means that the mean median mode are aligned too. When a data is skewed it indicates that the distribution is not a mirror image on both sides meaning that the data is tilted or spread out more towards either lower or higher values.

When the data is skewed your mean, median and mode appears to be a bit unbalanced as u can see in the picture below how those all are unbalanced in the below right skewed distribution.

RIGHT SKEWED DISTRIBUTION:

A right skewed distribution is a distribution where the mostly data lies in the left direction. Meaning that the distribution is properly being made on the left side and on the right side the data is skewed it could be due to many things such as outliers or anything else. Then the distribution is said to right skewed.



LEFT SKEWED DISTRIBUTION:

A left skewed distribution is a distribution where the mostly data lies in the right direction. Meaning that the distribution is properly being made on the right side and on the left side the data is skewed it could be due to many things such as outliers or anything else. Then the distribution is said to left skewed.

NOTE:

- Mean will always lie close to the tail.
- Mode will be there where most of data exist.
- Median will be in the center as usual.

If we need to calculate the central tendency of a skewed data we need to calculate median. But if its for normal distribution we will need to calculate mean first.

And if u are calculating measure of spread for skewed data we need to calculate the IQR and for normal distribution its better to calculate standard deviation first.

Z-SCORE:

Now we have already seen the normal distribution and they are very important in data as well as probability distribution.

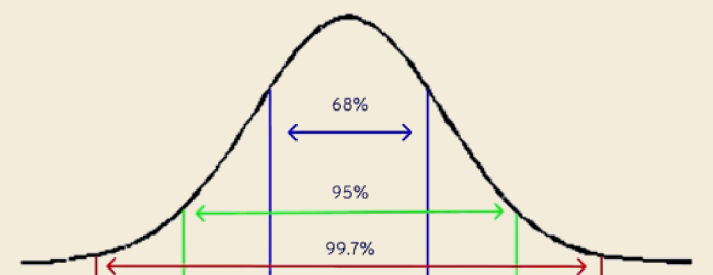
The area under the normal distribution is always 100% or 1.

(100% = 1, 99% = 0.9)



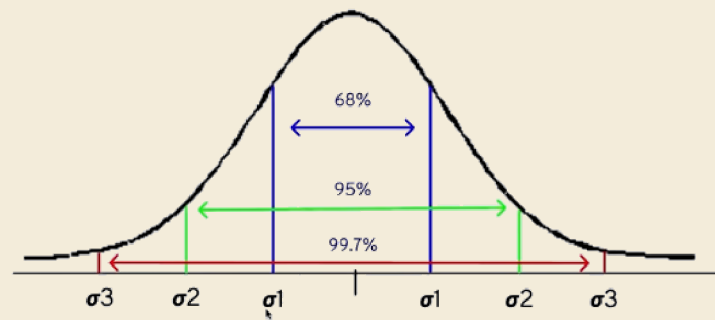
Now this normal distribution follows a formula known as Empirical Formula. Its basically a kind of rule that is followed in Normal distribution. The rule is simple 68-95-99.7 rule.

You might be thinking what are these numeric things here don't worry I gotchu.



We divide the distribution into 3 section following the Empirical Rule.

The percentage means probably this amount of data lies between these. Like if u see the blue arrows in between that there is a probability that 68% data lies similarly for green lines and arrows there is a probability that 95% of data lies here and for red arrows and line there is a probability that 99.7% data probability lies here.



Now a question might rise like how these lines are divided? The simple answer to this question is these lines are divided using standard deviation points.

Means that we will have a data and from that data we will be calculating the standard deviation points. Then we will be doing what is like draw lines from the standard deviation point 1 on the left side of mean to the right side of mean here we got our first point similarly a point after the first standard deviation point from the right side and left side here 95% data will lie then for the third one we will do the same. This all will be done when the distribution is normal and a bell shaped curve is formed.

These percentage means like the probability of getting 68% data lies here then 95% here and 99.7% lies here.

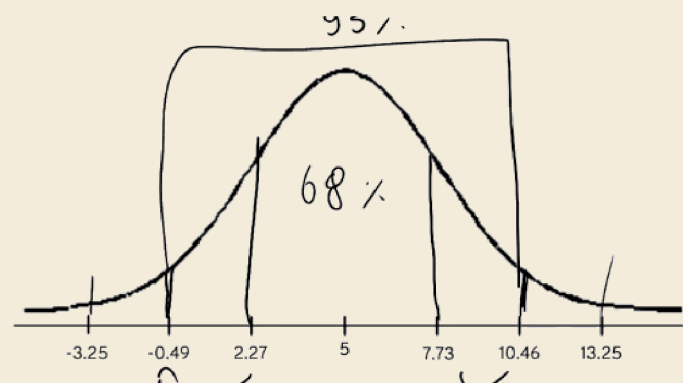
In short the probability of getting 68% data lies between std point 1 on both sides. 95% probability of getting data lies between std point 2 on both sides similarly for the 99.7% percent probability of getting data lies between std point 3 on both sides.

Now lets see this with an example of a consistent data

1,2,3,4,5,6,7,8,9

(Detailed calculation is done in the notebook)

Mean = 5



Standard Deviation = 2.73

Now for plotting we will keep 5 as center and subtract 5 from 2.73 for left side. And add 2.73 with 5 for right side. 5 will be center because its the mean.

5 will be subtracted or added then the new value that we will get in that value we will add or subtract the standard deviation.

In simple words this means that it is not necessary that the data lies between three standard deviation points in real life it may differ it could be due to an outlier which can cause things to behave differently. For proper understanding read all three attached pictures. In the above example we can clearly see all the data lies between 3 std points.

⚠ Important Point:

Even if your sample data shows all values within 3σ , we **don't generalize** that to the whole population. The **empirical rule** is about probability and the **shape of the distribution**, not just the data you have on hand.

Why Not 100% in Real Data?

When working with real datasets:

- You may observe that 100% of your data lies within 3σ .
- But that doesn't mean the population behaves that way.
- Real-world data may not be perfectly normal.
- Some datasets have no extreme outliers, giving you an impression that 3σ contains everything.

However, empirical rule is based on the ideal normal distribution, not your specific sample.

Key Reason: Theoretical vs. Actual Data

The 99.7% value comes from the mathematics of the normal distribution curve — it's a theoretical approximation. A normal distribution is continuous and unbounded, meaning:

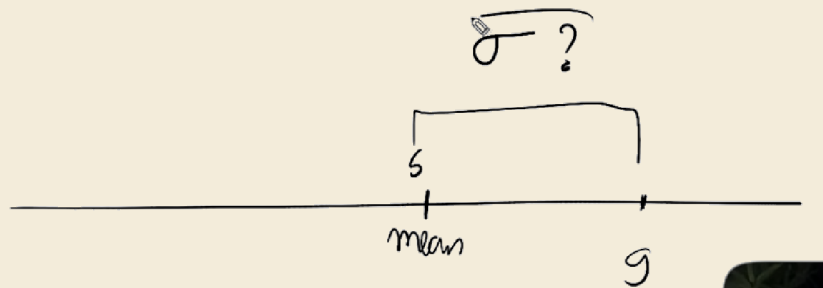
- It extends infinitely in both directions.
- There's always a tiny, non-zero probability that a data point lies beyond 3 standard deviations.

In theory:

- 100% of the data only lies within $\pm\infty$ standard deviations, not just ± 3 .
- Even past 3, 4, 5... standard deviations, there's still a very small probability.

NOW LET'S DISCUSS WHAT Z-SCORE IS :

A Z-Score is a powerful statistic tool which is used for measuring how far a data point is from the mean of a dataset, it is measured in terms of standard deviations.



From the attached example we have a data in which by using z-score we will measure how much far the data point 9 is from the mean. It is measured using standard deviation.

FORMULA OF Z-SCORE:

Z-Score = (data point – mean)/standard deviation

To understand this we will be using an example data point we have is 9.5 and mean for it is suppose it is 8 and standard deviation is 1 (usually 1 is not the standard deviation is using just for easy calculation). Now let's calculate the z-score

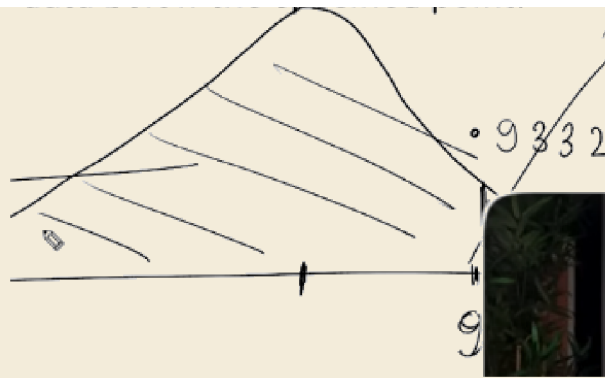
$$\text{Z-Score} = (9.5 - 8) / 1 = 1.5$$

This means 9.5 is 1.5 standard deviation away from the mean.

If the z-score value is positive then it means it's on the right side of the mean if the value is negative then it means it's on the left side of the mean.

Z-Score can also calculate the total percentage of data below a specific point. To do this we will use z-score table.

For seeing the z-table go to chrome and search z-table there will be multiple sites available. Coming towards our example first we calculate our z-score which was 1.5 and then we look at z-table we will search 1.5 since we have nothing after 1.5 we will look at 0 and the value for that is 0.9332.



This means that the data below that point holds 93.32% data. To determine how much data is above that point simply we will be subtracting $100 - 93.32 = 6.68\%$ data lies above this point

PROBABILITY

Think about probability like of an event occurring as a likelihood that it could occur.

Probability = Likelihood

EXAMPLE

For example like probability of rolling 6 on a dice.

Statistics and probability goes hand in hand because stats is about data and analysis of data.

Probability is going to allow us to determine how reliable our statistical result actually are.

So for that reason we will be studying probability and stats together.

Road-map of what we will study in Probability

Step 1 – Basics Probability Concepts

Step 2 – Types Of Events

Step 3 – Probability Rules

Step 4 – Conditional Probability

Step 5 – Bayes' Theorem (Important for ML Algorithm)

Lets start with probability. A probability always comes between 0 and 1. We can also tell it in percentage form or fraction form its same thing.

FORMULA:

Probability = True Outcome / Total Outcome

Lets take an example of tossing a coin and lets determine the probability of getting heads.

Total Outcome will be 2 as only 2 can come heads or tails. And the true outcome would be 1 because we can only get heads one time.

$$P(H) = \frac{1}{2}$$

$$P(T) = \frac{1}{2}$$

In short there is 50%-50% chance of getting a heads.

What is the probability of getting a 4 when we roll a dice?

$P(4) = \frac{1}{6} \Rightarrow \frac{1}{6}$ because we can only get 4 one times, while for 6 its the total outcome is 6 because we can get 6 occurrence when we roll a dice simple. SO the chance of getting 4 is 16.6%

TYPE OF EVENTS IN PROBABILITY

INDEPENDENT EVENT:

Those event that does not affect the occurrence of the other event. For example when we toss a coin and here comes heads if I will toss the coin again and tails comes this does not affect the occurrence of the previous event the probability for both the events will be

calculated differently.

Mathematically, two events **A** and **B** are independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Lets continue with the previous example

Let A be the event that the first toss is Heads.

Let B be the event that the first toss is Tails.

These two events are independent because they don't affect the occurrence of second event.

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

DEPENDENT EVENTS:

In probability dependent events are those events where the occurrence of one event affects the probability of the other.

Mathematically, two events **A** and **B** are dependent if:

$$P(A \cap B) \neq P(A) \cdot P(B)$$

or

$$P(B|A) \neq P(B)$$

LETS SEE AN EXAMPLE OF THIS:

EXAMPLE

Suppose we have 52 cards (I don't have much knowledge of these cards if I make mistake ignore it concept will be the same)

Let A be the event of drawing ace on the first draw.

Let B be the event of drawing ace on the second draw without putting the first card back.

Since the first card is not replaced, the total number of card is now 51. So the probability of B is affected due to A – they are dependent.

$$P(A) = \frac{4}{52}, P(B|A) = \frac{3}{51}$$

ACCORDING TO FORMULAE:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$= \frac{4}{52} \cdot \frac{3}{51}$$

$$= \frac{1}{221}$$

MUTUALLY EXCLUSIVE EVENT:

These are events which cannot happen at the same time. Two event A and B are mutually exclusive if:

$$P(A \cap B) = 0$$

That means that the probability of having these two events together is 0.

Lets see this event with an example:

EXAMPLE

Suppose u are rolling a dice and now lets find out can u roll 1 and 6 together this is impossible as you are rolling a single dice.

This is an example of this event.

(This event is not mostly asked even in hypothesis testing but still you have to know about it)

RULES OF PROBABILITY

ADDITION RULE:

The addition rule helps us to find the probability that at least one of two events occurs i.e. the probability of A or B happening.

In probability language $\rightarrow P(A \cup B) = P(A) + P(B)$

Whenever there is a question related to addition rule we mostly get choices between multiple things word like or, either and similar words that depict choices are present.

GENERAL ADDITION RULE (FOR ANY TWO EVENTS A AND B);

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$: Probability of A **or** B occurring
- $P(A \cap B)$: Probability that **both A and B** occur

We subtract $P(A \cap B)$ to **avoid double-counting** the overlap.

LETS UNDERSTAND WITH AN EXAMPLE:

What is the probability of drawing a red card or a king.

For instance there are 26 red cards in a deck of card and 4 kings among which 2 kings are red

$$P(A) = 26/52$$

26 means red cards and 52 are total cards

$$P(B) = 4/52$$

4 means king cards and 52 are total cards

$$P(A \cup B) = P(A) + P(B)$$

$$= 26/52 + 4/52$$

You might think that this is the answer it could be but here is a catch in king cards there are 2 black and 2 red cards in the p(b) we took all 4 cards as a king while 2 red kings are already in the red card probability so properly we will be writing it as

$$26/52 + 4/52 - 2/52$$

subtracting 2 red kings as they are already in the red card

answer will be 7/13

EXAMPLE

Example of this scenario is as follows:

SPECIAL CASE: MUTUALLY EXCLUSIVE EVENTS

If A and B are **mutually exclusive** (they cannot happen together), then:

$$P(A \cup B) = P(A) + P(B)$$

Rolling an even number in a fair six sided dice

Rolling an odd number in a fair six sided dice

$$P(A) = 3/6 \rightarrow (3 \text{ because only 3 even numbers can actually come})$$

$$P(B) = 3/6 \rightarrow (3 \text{ because only 3 odd numbers can actually come})$$

$$P(A \cup B) = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} = 1 \text{ Simple answer}$$

MULTIPLICATION RULE:

Multiplication Rule is used to find the probability of two events happening together i.e. the probability of A and B occurring.

In probability language $P(A \cap B)$

FOR INDEPENDENT EVENTS:

If A and B are independent then

$$P(A \cap B) = P(A) \cdot P(B)$$

We use this when one event does not affect the another event.

FOR DEPENDENT EVENTS:

If A and B are dependent then

$$P(A \cap B) = P(A) \cdot P(B|A)$$

Use this when the probability of B depends on whether A has occurred.

For better understanding we will see an example like drawing a king for two times.

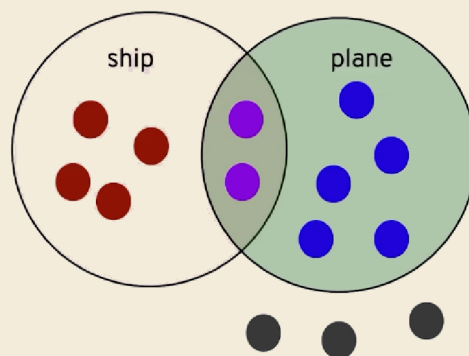
$$P(A) = 4/52$$

and for drawing the king again the probability will be

$$P(B) = 3/51$$

because a king is already drawn before.

Conditional Probability



EXAMPLE

Suppose this is an example of the conditional probability we can see that approx 4 people loves ship, 5 people loves ship, 2 people loves both of them and 3 people don't like anything.

(This is an independent event)

For better understanding and solving these type of questions we should prepare a contingency table.

Now lets calculate their probability →

	loves ship	no ship
love plane	2	5
no plane	4	3

	loves ship	no ship
love plane	2 p - 2/14	5 p - 5/14
no plane	4 p - 4/14	3 p - 3/14

(For better understanding look in notebook their calculations have been done properly and visually too.)

Now we will calculate the total of rows and columns too

	loves ship	no ship	row total
love plane	2 p - 2/14	5 p - 5/14	2 + 5 p - 7/14
no plane	4 p - 4/14	3 p - 3/14	4 + 3 7/14
column total	2 + 4 p - 6/14	5 + 3 p - 8/14	

Now we will see what type of question we get

QUESTION

A Person who loves a plane what is the probability of him loving ship too

There are total of 7 person who loves plane out of them only 2 loves ship too. The probability will be 2/7. We solved this logically let see the formula approach too.

FORMULA

Formula $\rightarrow P(A|B) = P(A)/P(B)$

$P(A) \rightarrow$ Probability of a person who loves plane and ship

$P(B) \rightarrow$ Probability of a person who loves plane

$$P(A|B) = 2/14 / 7/14 = 2/7$$

Condition probability are also used in Dependent Events its simple too the formula differ

FORMULA

Formula $\rightarrow P(A \cap B) = P(A) * P(B|A)$

Lets see an example there are 3 white balls and 2 red balls what is the probability of getting 2 white balls

ITS SIMPLE:

$$P(A \cap B) = 3/5 * 2/4 = 3/10$$

$P(A)$ here means the total of white balls we have and $P(B)$ here means the probability of getting 2 white balls we used 2/4 because we have to calculate the probability of getting 2 white balls so we subtracted 1 ball from the total and white ball too.

QUESTION

Questions related to probability are done in the notebook we have done Questions related to Probability, Rules of Probability, Conditional Probability in the notebook.

A die is rolled. What is the probability of getting an even number or a number less than 3?

A card is drawn from a deck. What is the probability of getting a king or a red card?

More Questions done in the notebook.

In a class of 40 students, 18 like Maths, 24 like Science, and 10 like both. What is the probability that a student chosen at random likes either Maths or Science?

A bag contains **4 red** and **3 blue** marbles. Two marbles are drawn **without replacement**.
What is the probability that both marbles are red?

BAYES' THEOREM

Bayes' Theorem is a probability theorem that helps us find the probability of an event or cause after observing some evidence that support your result.

It is used to determine the most likely cause of an observed event by updating our prior belief using new information.

EXAMPLE:

Let's break it down:

- **A = Banda corona positive hai**
- **B = Banda kisi corona wale ke contact mein aaya th**

Aapko mila hai **B (evidence)**, aur aap soch rahe ho:

“ *"B milne ke baad, A hone ka chance kitna hai?"* ”

Yehi to hai: BAYES theorem

$$P(\text{Corona}|\text{Contact}) = \frac{P(\text{Contact}|\text{Corona}) \cdot P(\text{Corona})}{P(\text{Contact})}$$

FORMULAE:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

The questions related to this theorem are very big to solve these we need to do case theories but here we are solving a basic question (Solved in notebook):

Lets say we have 2 dice one dice is fair and another dice is weighted as the chance of getting a 6 on that dice is 50%.

1-Fair Dice

2-Unfair Dice

Scenario → We choose one of the dice without knowing which die we picked

We roll the dice and got a 6/

Now calculate the probability that we picked the biased dice given that we rolled a 6

To solve this we are going to break things as follows:

A-Choosing Biased Die

B-Rolling a Dice

So

$$p(A|B) \rightarrow p(\text{Biased}|6)$$

$$p(B|A) \rightarrow p(6|\text{Biased}) = 1/2 \text{ (reason in the notebook)}$$

$$p(A) \rightarrow p(\text{Biased}) = 1/2$$

$$p(B) \rightarrow p(6) = p(\text{biased} * \text{rolling } 6) + p(\text{fair} * \text{rolling } 6)$$

$$= 1/2 * 1/2 + 1/2 * 1/6$$

$$= 1/3$$

Lets put the value in the formulae:

$$p(A|B) = \frac{1}{2} * \frac{1/2}{1/3} = \frac{3}{4} \text{ answer (proper calculation is done in the notebook with reason and working refer to that)}$$

“ Bayes' Theorem batata hai ki mere intuition ko nayi evidence milne ke baad kitna support milta hai.”

Ya aur casually:

“ "Bayes' Theorem = Mere gut feeling + Maths se confirmation!" 😊

Before we begin with hypothesis testing lets understand the difference between descriptive and inferential statistics

DESCRIPTIVE VS INFERENTIAL

	Descriptive	Inferential
Purpose	Summarize & describe data	Make predictions or draw conclusions from data
Works with	Actual data you collected	Goes beyond the data to generalize to a population
Examples	Mean, Median, Mode, Range, Standard Deviation, Graphs	Hypothesis Testing, Confidence Intervals, p-values, t-tests
Focus	"What does the data say?"	"What can we infer about the population?"

Let's say you survey **100 students** on their exam scores.

- **Descriptive Stats:**
"The average score is 72.3, and the standard deviation is 8.5."
→ You're just **describing what you observed**.
- **Inferential Stats:**
"We are 95% confident that the average score of **all students in the school** is between 70 and 75."
→ You're **inferring** something about a **larger population** based on your sample.
- We have already covered descriptive stats and now it's time to start with inferential stats.

Inferential Stats

It is the branch of statistics that helps us to make predictions or decisions about a population based on a sample of data.

Hypothesis Testing

A hypothesis testing is a statistical method used to decide whether there's enough evidence in sample to support a claim about a population.

Lets see with an example how hypothesis testing works:

I have a coin and I want to test whether a coin is fair or not by flipping in 100 time.

$$p(h) = 0.5 \text{ and } p(t) = 0.5$$

If we think logically we can assume that if a coin is flipped 100 times 50 times head will come and 50 times tails will come but sometimes this will not be the case.

To test that our coin is fair we will be using hypothesis testing and for hypothesis testing we have to define 2 hypothesis null and alternate.

Null Hypothesis → Coin is fair.

Alternate Hypothesis → Coin is unfair.

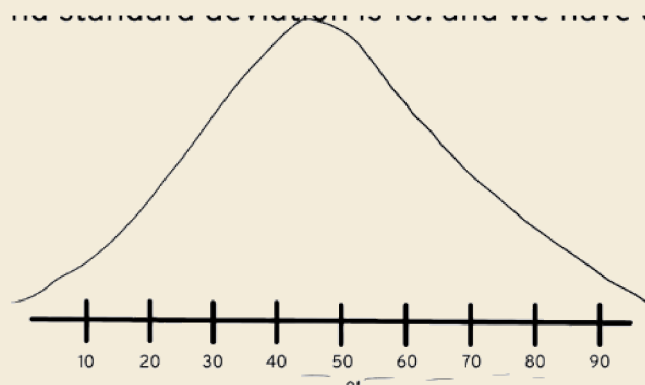
Null hypothesis is always seen the positive side because there is a quote 'Innocent Until Proven Guilty'.

Alternate Hypothesis is just the opposite side of Null Hypothesis.

The third thing we will do is do an experiment.

After experiment we will Accept or reject the null hypothesis.

For this question lets say before doing the experiment we got the information that our mean is 50 and standard deviation is 10 and the distribution is normal. Now we can create a bell curve over the plot just like this.



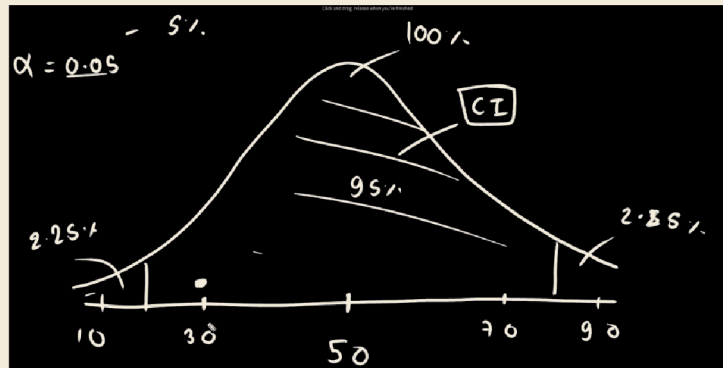
Now suppose I performed the experiment and I got heads 30 times. So can we say that our coin is fair or not? So to see whether our coin was fair or not our experiment head count must be close to mean but how much close for this we will use an important concept of significance value.

Significance value is set by the domain experts and lets say the significance value is 0.05 and it is denoted by α → Alpha sign. And using this significance value we calculate the confidence interval

Confidence interval = $1 - \text{Significance value} = 1 - 0.05 = 0.95$.

Confidence interval means u are in the zone where u can determine that your coin is fair

(Domain experts are those who work in stats and create theories and much more. Usually there are mostly 3 values for significance value 0.10, 0.05, 0.01. If the value is not given we use 0.05 it is denoted by α .)



Looking at the image that I extracted from the video. We can see the above example properly drawn the two tails thing is something which is covered in the upcoming topic too. The two tails were made by dividing the significance value by 2 splitting the data into two tails and the remaining part is the confidence interval that means if the data lies between those lines we can be confident and choose our null hypothesis and if the values lie outside of the tail we can clearly say we reject our Null hypothesis.

Further details of this will be covered in the upcoming topic.

TYPE 1 AND TYPE 2 ERROR:

In this scenario, the null hypothesis () is that a person does not have COVID-19, and the alternative hypothesis () is that the person has COVID-19.

- **Type I Error (False Positive):** This occurs when the test result indicates the person has COVID-19, but they are actually healthy. The speaker notes this is like falsely telling a healthy person they are sick.
- **Type II Error (False Negative):** This occurs when the test result indicates the person does not have COVID-19, but they are actually infected. The speaker describes this as a more dangerous error, as a sick person is told they are healthy and may continue to spread the virus.

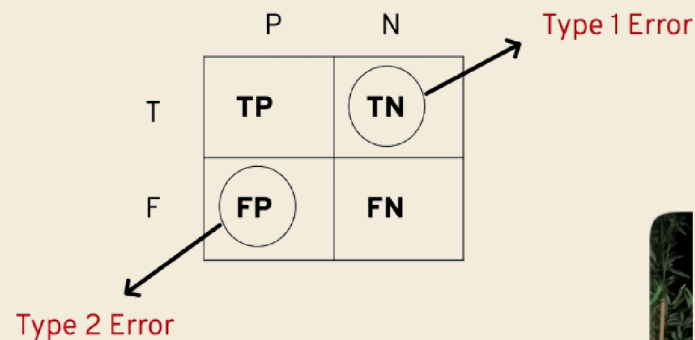
CONFUSION MATRIX:**TYPE 1 AND TYPE 2 ERROR**

So based on this information we will have 4 outcomes :

- 1) We reject the null hypothesis in reality it was false.
- 2) We reject the null hypothesis in reality it was true. **Type 1 error**
- 3) We accept the null hypothesis in reality it was True.
- 4) We accept the null hypothesis in reality it was false. **Type 2 error**

NOTE

An important concept in machine learning based on all the outcomes we can create like this

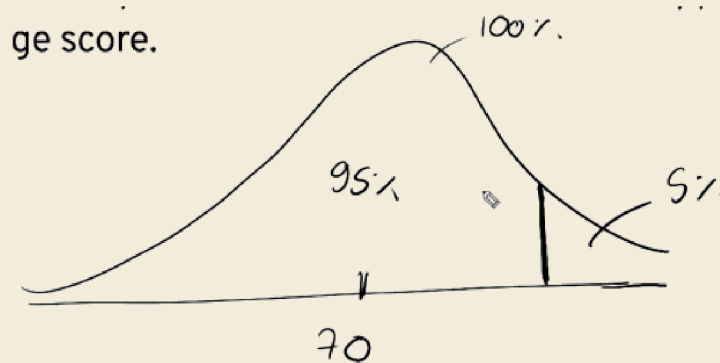
**ONE TAILED AND TWO TAILED TEST:**

This is the one tailed and two tailed thing we were talking about before to determine how many tails one or two we have to use for the test lets understand with an example You have created an application and u claim that students who study through your application scores on an average of 70 marks in a particular subject.

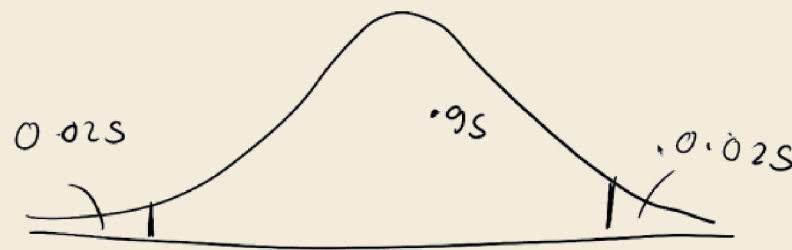
Now u want to verify this claim in statistics You collect a sample of 30 students who uses ur application to verify.

Since u have claimed that students who use ur application score 70 marks and haven't given any specific range we can clearly see we will be using a single tail because there is only a single claim. The significance value we had will only implies with one tail no dividing them and splitting them.

The 5% data left will go in the tail. While the other 95% data will remain in the center.



Coming towards the 2 tail thing we will be using the same example just twisting it like before we have used an example of students scoring 70 marks in subject here we will be making it like different than 70 it will mean above or below 70. Now we have a thing like above or below we will be using two tails like dividing the significance value with 2 so distributing them equally on 2 halves. Just like the above image we are seeing the 95% data will lie in the center while the 2.5% and 2.5% of data lies outside of both the tails. I hope u got my point if still didn't got my point look at the video we will properly learn this once we solve these questions.



Z-TEST:

Before we begin with this let's first see how different things are written in population and sample we have discussed it before too but for instance see it below:

We will use z-test when the sample is greater than 30 and you must also know the variance or standard deviation of population too.

	Population	Sample
mean	μ	$\bar{x}$
Variance	σ^2	s^2
STP	σ	s
Count	N	n

Q. What does z-test check?

A. It checks if the sample mean significantly different from the population mean.

Z-TEST FORMULA:

- $\bar{x}$ = Sample mean
- μ = Population mean
- σ = Population standard deviation
- n = Sample size

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

where →

Before beginning with this we need to setup Null Hypothesis and Alternate Hypothesis

QUESTION

Sample Question

Mean = 74

Population Mean = 70

Population Standard Deviation = 10

Sample Size = 36

Setting up two essential things

Null Hypothesis → No difference

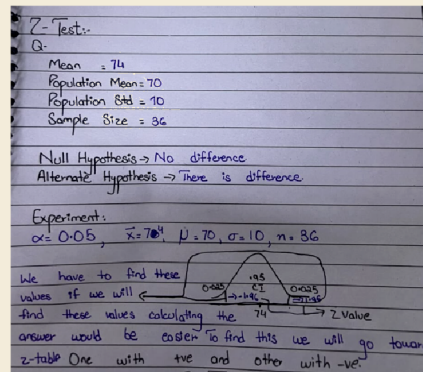
Alternate Hypothesis → There is a difference

Then we will perform an experiment to perform an experiment we need to assume the significance value as no significance value is given we will be assuming the significance value as 0.05.

A z-test is performed between elements where there is a normal distribution.

Here from the question it was obvious that we will be using two tail test because we need to calculate the difference between the means it was obvious we will be using two tail test.

THE FIRST WAY TO SOLVE Z-TEST:



$$= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$= \frac{74 - 70}{10 / \sqrt{36}} = 2.4$$

Since the value is higher than the critical value. This means we are rejecting the Null Hypothesis as our values don't lie under confidence interval.

The second way is to solve using the p-values which is used when we are solving through coding approach while for the normal approach like doing on pen and paper we use critical points approach. For now we will be solving more questions related to critical value which was our first way to solve it.

For seeing the values in z-table when we are using two tails the approach is simple divide the significance value with 2 and distribute in two halves each on left right sides then to find the z-table value u have to do is 1-right tale value and the answer that will come that number we have to look for that or a number close to that in the table u will get a point that will be the critical value like for the above question we had 0.9950 we look in the table it was in 2.5 and in that row it lied in 0.7 so ultimately we got our critical point 2.57 or 2.58 simple is that this approach is for two tails. We haven't done for left or right tail till yet I will add an explanation for that too once we reach there.

The average marks of Sheryians students are 80 with the population standard deviation of 2.5 but a teacher believes the mean to be different so he took the marks of 36 individuals and found the average to be 82.5

A health researcher claims that the average weight of adult males in a city is 70 kg.

A sample of 49 men is taken, and their average weight is found to be 72 kg.

Assume population standard deviation (σ) is 8 kg.

👉 At 5% significance level, **test the claim** using a one-sample Z-test.

A company claims that its new software bug fix takes **no more than 30 minutes** on average.

A sample of 36 fixes shows an average time of 32 minutes.
Population standard deviation is known to be 6 minutes.

👉 Test the company's claim at the 0.01 significance level.

One tail test

It is believed that the average score in a statistics exam is **75**.

A professor thinks students this year have performed **worse**.

A random sample of 40 students shows an average score of 72,
with a population standard deviation of 10.

👉 Test the professor's belief at the 0.05 significance level.

When we calculate z-value instead of using critical values we use p-values which is known as percentage values or probability values. We have done this in code refer to that if u are here looking for code here there is only theoretical part no code here ;) for codes look at the repository in the same folder u find this PDF u will get the codes too in the same folder.

These values are used to find the area inside of the z-score. We will be using these values to compare them.

FORMULA FOR P-VALUE:

```
p_value = 2 * (1-norm.cdf(abs(z_score)))
```

Before we begin with T-Test make sure to solve all the question in the other notes given by the instructor.

T-TEST:

We use T-Test when we have an unknown standard deviation of population.

Sample size it can be smaller than 30 or even above then 30 but usually its smaller.

FORMULA:

Its same as Z-Test it only has sample standard deviation instead of population standard deviation.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

In this test we also need to calculate Degree Of Freedom and the formula is as follows for Degree Of Freedom.

$Df = n-1$

we will understand what df is when we will see T-Table.

Now lets solve a example of how T-Test works.

For finding the critical value we will go on t table website and look the value there since this question clearly says difference its obvious we will be using two tails we will go to two tails t-table we will see the value of 0.05/0/025 in the 9 df row since our df is 9 and we will fetch that value.

A bakery claims that their cupcakes weigh **at least 150 grams** on average.

A customer suspects this isn't true.

They take a random sample of 10 cupcakes and find:

- Sample mean = **145 grams**
- Sample standard deviation = **8 grams**

👉 At a 5% significance level, test the customer's claim using a one-sample **T-Test**.

Solution is done in the notebook refer to that.

A nutritionist wants to verify if a new diet plan helps people lose **more than 5 kg** in a month. From a sample of 15 people, the **average weight loss** was **5.6 kg**, with a **standard deviation of 1.2 kg**.

Test the hypothesis at a 5% significance level. Can we conclude the diet helps reduce more than 5 kg?

Q2 also done in notebook

T-Test is more used in coding and daily life.

We have done p-value approach to look for code in the repository.

TWO SAMPLE TEST:

We use two sample test whether its z-test or t-test when:

We have two samples from two different groups and we have to compare their means to see if there is a statistically significant difference between them.

LETS SEE AN EXAMPLE:

We have a comparison based data

Group1: Students from School A

Group2: Students from School B

We want to see if School A performs better than School B students on average

Before we begin with this lets see their formulas first for calculating two sample test we have these formulas.

If there is a scenario like population's std is given not the sample one so we will use z-test.

Formula for
two sample Z-Test

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$



Formula for
two sample T-Test

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \min(n_1-1, n_2-1)$$

For t-test if we have the std of sample is given not the population one so we will use t-test.

When choosing between the type of test our main focus should be on the type of standard deviation provided.

👉 If the **sample standard deviation** is given (and **population standard deviation is unknown**), you should use the **T-test** – even if the sample size is greater than 30.

A teacher wants to know if there's a difference in math scores between boys and girls.

- Boys: $n = 30$, mean = 70, std dev = 8
- Girls: $n = 28$, mean = 74, std dev = 6

After seeing the above question we can clearly see like we have been given the standard deviation of sample and for that we will be clearly using the t-test.

Mostly we are given the std of sample so we will be solving using t-test.

(Solve more questions related to this test if needed)

We are now moving towards code solving it using p-values instead of critical values.

CHI SQUARE TEST:

Chi Square Test is used for categorical values it is used to check if there is a significant association between two categorical variables. It helps to answer questions like:

- Is there a relationship between gender and voting preference?

- Do two different marketing strategies results in different customer preference?
- Is there a difference between observed and expected frequencies in a survey or experiment?

The formula for Chi Square Test is as follows:

where

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

O = Observed Frequency (actual data)

E = Expected Frequency (what we expect under the null hypothesis)

Σ = Sum over all categories

STEPS ARE SIMILAR:

STATE THE HYPOTHESIS:

- Null Hypothesis: There is no association (or no difference) between the variables.
- Alternate Hypothesis: There is an association (or a difference)
- Calculate Expected Frequencies: For each cell in your data the expected frequency is calculated using:

$E = (\text{Row total} * \text{Column total}) / \text{Grand total}$

- Calculate The Chi Square statistic: Apply the formula
- Determine The Degree Of Freedom: For a test of independence:

$df = (r-1)(c-1)$

- Where r = number of rows and c = number of columns

Lets see with an example solution is done in notebook

STEPS:

Let's say we want to check if there's a relationship between **Gender** and **Choice of Drink**. We survey 100 people, and the data is:

	Coffee	Tea	Juice	Total
Male	30	10	5	45
Female	15	20	20	55
Total	45	30	25	100

We will calculate the expected frequency

Then apply the formula after that we will find degree of freedom then we will look for the significance value based on df value we will get in the chi square table properly solved and explanation is in the notebook.

CHI SQUARE TEST:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where
 O_i = Observed Frequency
 E_i = Expected Frequency
 Σ = Sum over all categories

$E_i = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$

Q. Gender

	Coffee	Tea	Juice	Total
Male	30	10	5	45
Female	15	20	20	55
Total	45	30	25	100

Expected Table

Gender	Coffee	Tea	Juice	Total
Male	$\frac{45 \times 45}{100} = 20.25$	$\frac{45 \times 30}{100} = 13.5$	$\frac{45 \times 25}{100} = 11.25$	45
Female	$\frac{55 \times 45}{100} = 24.75$	$\frac{55 \times 30}{100} = 16.5$	$\frac{55 \times 25}{100} = 13.75$	55

Gender	Drink	Observed	Expected
Male	Coffee	30	24.25
Male	Tea	10	15.75
Male	Juice	5	11.25
Female	Coffee	15	24.75
Female	Tea	20	16.25
Female	Juice	20	19.75

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$= \frac{(30-24.25)^2}{24.25} + \frac{(10-15.75)^2}{15.75} + \frac{(5-11.25)^2}{11.25} + \frac{(15-24.75)^2}{24.75} + \frac{(20-16.25)^2}{16.25} + \frac{(20-19.75)^2}{19.75}$$

$$= 16.49$$

$$df = (r-1) \times (c-1) = (2-1) \times (3-1) = 2$$

Now look for critical value at Chi Table at 0.05

Critical value at $\alpha = 0.50$ at df value 2 is 5.991

Based on this we reject Null Hypothesis

Code implemented for chi square too look in the repository.

ANNOVA TEST:

ANNOVA (Analysis of Variance) is used to compare means of more than two independent groups to check if at least one group mean is significantly different.

Lets see an example suppose we are comparing the average math test scores of students from three different teaching methods (Group A, Group B, Group C). We want to know Do these teaching methods result in different average scores?

(We will not be solving this manually because it would be time taking and very much complex we are directly moving towards the code part as said by the instructor)

CO VARIANCE:

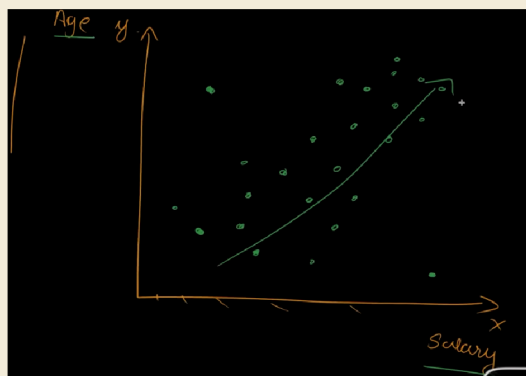
Covariance tell how 2 variable changes together. Covariance is used majorly in machine learning but its a part of statistics.

Now if you don't remember what are variables.

Reminder when we have a table and in that table we want to see the connection between 2 columns we use covariance.

	Age	Salary
	22	26k
	23	34k
	24	40k
	25	45k
	26	50k

This is a scatter plot of age and salary to see the relationship between these both by using this plot we create linear regression algorithms. The arrows shows the direction of relationship.



A co relationship gives result in form of positive and negative. If the result is positive that means that the relationship is positive if the result is negative that means the relationship is negative.

FORMULA IS AS FOLLOWS:

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

$\bar{X}$ is the mean of first var and x means elements in table. $\bar{Y}$ is the mean of second var and y means elements in that table.

To find the coorelation we have to find covariance first and then divide it by std of both the variables

So what covariance tells us basically it just tells us the direction the relation is positive or negative but to actually see the strength of relation we have to use the coorelation.

(Solution done in notebook) answers are as follows:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

So with this data our covariance was- 14.75

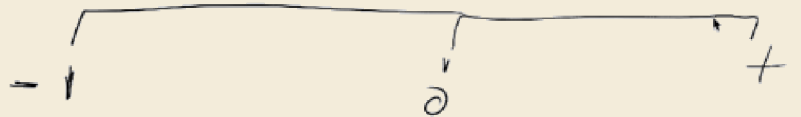
Now we just have to divide it by the standard deviation of both variables.

Age $\sigma = 1.58$

Salary $\sigma = 9.38$

correlation = $14.75 / 14.82 = 0.99$

When we talk about coorelation we will get the result between -1 and +1 . The more close it is like for the above one we have 0.99 this means its strongly related to each other if it would be -0.99 this means they aren't related to each other. This is used in machine learning when we want to predict salary of employees.



(Stats completed)